

MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS
NANYANG TECHNOLOGICAL UNIVERSITY

Tutorial 6

HINTS

We will discuss several questions from Chapter 5 in Ross' book.

Tutorial Questions

- (1) Two types of coins are produced at a factory: a fair coin and a biased one that comes up heads 55 percent of the time. We have one of these coins, but do not know whether it is a fair coin or a biased one. In order to ascertain which type of coin we have, we shall perform the following statistical test: We shall toss the coin 1000 times. If the coin lands on heads 525 or more times, then we shall conclude that it is a biased coin, whereas if it lands on heads less than 525 times, then we shall conclude that it is a fair coin. If the coin is actually fair, what is the probability that we shall reach a false conclusion? What would it be if the coin were biased?

Answer: When the coin is fair, the number of heads X follows $\text{Bin}(1000, 1/2)$, and so $\mathbb{E}(X) = 500$, $\text{Var}(X) = 250$. Thus

$$\begin{aligned} P\{X \geq 525\} &= P\{X \geq 524.5\} \\ &= P_{Z \sim N(0,1)} \left\{ Z \geq \frac{524.5 - 500}{\sqrt{250}} \right\} \\ &= P_{Z \sim N(0,1)} \{Z \geq 1.55\} \\ &= 1 - \Phi(1.55) \\ &\approx 1 - 0.9394 \\ &= 0.0606. \end{aligned}$$

When the coin is biased, the number of heads Y follows $\text{Bin}(1000, 0.55)$, then $\mathbb{E}(Y) = 550$ and $\text{Var}(Y) = 247.5$. Thus

$$\begin{aligned} P\{Y < 525\} &= P\{Y < 524.5\} = P_{Z \sim N(0,1)} \left\{ Z \leq \frac{524.5 - 550}{\sqrt{247.5}} \right\} \\ &= P_{Z \sim N(0,1)} \{Z \leq -1.62\} \\ &= 1 - \Phi(1.62) \\ &\approx 1 - 0.9474 \\ &= 0.0526. \end{aligned}$$

(Remark: The exact answers are $0.060607 \dots$ and $0.052681 \dots$).

- (2) (a) A fire station is to be located along a road of length A , $A < \infty$. If fires occur at points uniformly chosen on $(0, A)$, where should the station be located so as to minimize the expected distance from the fire? That is, choose a so as to

$$\text{minimize } \mathbb{E}(|X - a|)$$

when X is uniformly distributed over $(0, A)$.

- (b) Now suppose that the road is of infinite length—stretching from point 0 outward to ∞ . If the distance of a fire from point 0 is exponentially distributed with rate λ , where should the fire station now be located? That is, we want to minimize $\mathbb{E}(|X - a|)$, where X is now exponential with rate λ .

Answer:

- (a) As X denotes the distance from the place where fire happens to the beginning point, and X is a uniform distribution on $(0, 1)$, we have Then

$$\begin{aligned}\mathbb{E}(|X - a|) &= \int_0^A \frac{1}{A} |x - a| dx = \frac{1}{A} \int_0^a (a - x) dx + \frac{1}{A} \int_a^A (x - a) dx \\ &= \left(a - \frac{1}{2}A\right)^2 + \frac{1}{4}A^2.\end{aligned}$$

$\mathbb{E}(|X - a|)$ can have minimum when $a = 1/2A$.

- (b) Again, X denotes the distance from the place where fire happens to the beginning point, and X is a exponential distribution on $(0, \infty)$ with parameter λ . Then

$$\begin{aligned}\mathbb{E}(|X - a|) &= \int_0^\infty |x - a| (\lambda e^{-\lambda x}) dx \\ &= \int_0^a (a - x) (\lambda e^{-\lambda x}) dx + \int_a^\infty (x - a) (\lambda e^{-\lambda x}) dx \\ &= a + \frac{2}{\lambda} e^{-\lambda a} - \frac{1}{\lambda}.\end{aligned}$$

Differentiating w.r.t. a and letting the derivative be 0, we obtain that

$$1 - 2e^{-\lambda a} = 0, \quad a = \frac{\ln 2}{\lambda}.$$

Thus $\mathbb{E}(|X - a|)$ has minimum when the fire station is $\frac{\ln 2}{\lambda}$ from the beginning of the road.

- (3) Suppose that the length of a phone call in minutes is an exponential random variable with parameter $\lambda = 1/10$. If someone arrives immediately ahead of you at a public telephone booth, find the probability that you will have to wait
- (a) more than 10 minutes;
 - (b) between 10 and 20 minutes.

Answer: Let X denote the length of the call made by the person in the booth. Then the desired probabilities are

- (a) $P\{X > 10\} = 1 - F(10) = e^{-\frac{1}{10} \times 10} = e^{-1} \approx 0.368$.
- (b) $P\{10 < X < 20\} = F(20) - F(10) = e^{-1} - e^{-2} \approx 0.233$.

- (4) If X is an exponential random variable with parameter $\lambda = 1$, compute the probability density function of the random variable Y defined by $Y = \ln X$.

Answer: The CDF of Y is

$$P\{Y \leq y\} = P\{\ln X \leq y\} = P\{X \leq e^y\} = \int_0^{e^y} e^{-x} dx = [-e^{-x}]_0^{e^y} = 1 - e^{-e^y}.$$

Differentiating to obtain the density function of Y ,

$$p_Y(y) = \frac{d}{dy} P\{Y \leq y\} = (1 - e^{-e^y})' = -e^{-e^y} (-e^y) = e^{y-e^y}, \quad -\infty < y < \infty.$$

(5) If X has a probability density f_X . Find the density function of $Y = |X|$.

Answer: It is true that for $y < 0$, $f_Y(y) = 0$, as $Y = |X|$ never takes negative values.

For $y \geq 0$,

$$F_Y(y) = P\{Y \leq y\} = P\{|X| \leq y\} = P\{-y \leq X \leq y\} = F_X(y) - F_X(-y),$$

and then, by differentiation,

$$f_Y(y) = f_X(y) - (f_X(-y))(-y)' = f_X(y) + f_X(-y).$$

(6) Let

$$f(x) = \frac{\theta^2}{1+\theta}(1+x)e^{-\theta x}, \quad x > 0$$

where $\theta > 0$.

(a) Show that f is a probability density function.

(b) If X has density function f , find $\mathbb{E}(X)$ and $Var(X)$.

Answer: (a) It is obvious that f is nonnegative everywhere. We show that

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

by

$$\begin{aligned} \int_{-\infty}^{\infty} f(x)dx &= \int_0^{\infty} \frac{\theta^2}{1+\theta}(1+x)e^{-\theta x}dx = \frac{\theta}{1+\theta} \int_0^{\infty} (1+x)\theta e^{-\theta x}dx \\ &= \frac{\theta}{1+\theta} \left[\left(-(1+x)e^{-\theta x} \right)_0^{\infty} + \int_0^{\infty} \theta e^{-\theta x}dx \right] = \frac{\theta}{1+\theta} \left[1 + \frac{1}{\theta} \right] = 1. \end{aligned}$$

(b)

$$\begin{aligned} \mathbb{E}(X) &= \int_{-\infty}^{\infty} xf(x)dx = \int_0^{\infty} \left(x \frac{\theta^2}{1+\theta}(1+x)e^{-\theta x} \right) dx \\ &= \frac{\theta}{1+\theta} \int_0^{\infty} x\theta(1+x)e^{-\theta x}dx = \frac{\theta}{1+\theta} \int_0^{\infty} (x+x^2)d(-e^{-\theta x}) \\ &= \frac{\theta}{1+\theta} \left[\left(x(-e^{-\theta x}) \right)_0^{\infty} + \int_0^{\infty} e^{-\theta x}dx + \left(x^2(-e^{-\theta x}) \right)_0^{\infty} + 2 \int_0^{\infty} x(e^{-\theta x})dx \right] \\ &= \frac{\theta}{1+\theta} \left[0 - \frac{1}{\theta} \left(e^{-\theta x} \right)_0^{\infty} + 0 + 2 \frac{1}{\theta} \int_0^{\infty} xd(-e^{-\theta x}) \right] \\ &= \frac{\theta}{1+\theta} \left[\frac{1}{\theta} + \frac{2}{\theta} \left(\left(x(-e^{-\theta x}) \right)_0^{\infty} + \int_0^{\infty} e^{-\theta x}dx \right) \right] = \frac{\theta}{1+\theta} \left[\frac{1}{\theta} + \frac{2}{\theta^2} \right] = \frac{\theta+2}{\theta^2+\theta}. \end{aligned}$$

To obtain $Var(X)$, we first calculate $\mathbb{E}(X^2)$:

$$\mathbb{E}(X^2) = \int_{-\infty}^{\infty} x^2 f(x)dx = \int_0^{\infty} \left(x^2 \frac{\theta^2}{1+\theta}(1+x)e^{-\theta x} \right) dx,$$

as above, we can have

$$\mathbb{E}(X^2) = \frac{\theta}{1+\theta} \left(\frac{2}{\theta^2} + \frac{6}{\theta^3} \right).$$

This gives

$$Var(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \frac{\theta}{1+\theta} \left(\frac{2}{\theta^2} + \frac{6}{\theta^3} \right) - \left(\frac{2+\theta}{\theta(1+\theta)} \right)^2.$$